

Anshay Agarwal

Senior AI Engineer · Medical imaging, biosignals and real-time voice AI

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India · open to remote work and relocation · available to join immediately

SUMMARY

Senior AI engineer who has built imaging, signal-processing and AI systems since 2013, on real hardware and real clinical data, at Qualcomm, Nvidia and India's defence research organisation (DRDO). I build applied AI end to end: medical-image and biosignal pipelines, phone-based health screening and real-time voice agents, from first prototype to production.

HEALTH AND BIOSIGNAL AI

Vitalis — health screening on an ordinary phone

2026

Flutter, FastAPI, Cloud Run GPU, MedGemma, HeAR, MedSigLIP

- Seven screening checks on a phone, plus a medical chat assistant on **MedGemma 27B** (multimodal). Built as a screening aid: it flags what is worth showing a clinician and never diagnoses.
- **Pulse from the front camera, fully on device** (remote photoplethysmography): FaceMesh cheek tracking, POS colour projection and an FFT limited to 42–240 BPM, computed at the camera's measured frame rate rather than an assumed 30 fps.
- **Tremor screening from the accelerometer, fully on device**: 3–12 Hz analysis split into Parkinsonian (3–6.5 Hz) and essential-tremor (6.5–12 Hz) bands, reported in m/s² so results compare across phones.
- Cough and voice checks on Google's **HeAR** acoustic model; jaundice and oral-lesion checks on **MedSigLIP** image embeddings; an anaemia check from an inner-eyelid photo.

Artery segmentation in CT angiography (consulting project)

2022 – 2023

Python, PyTorch, 3D U-Net, AWS

- Built a **3D segmentation pipeline** that isolates the carotid and vertebral arteries in **CT angiography** scans of real patients.
- Added calcification classification to measure arterial plaque burden, as an input to automated **stroke-risk assessment**.
- In the same engagement, built a CNN that predicts the effect of **protein mutations**, and made the AWS ML pipeline **30–50% faster**.

NeurIPS 2026 EEG Foundation Challenge — cross-session EEG decoding (Track 2)

2026, final phase to Nov 2026

Python, PyTorch, braindecode, pyRiemann

- Entry for decoding EEG from later recording sessions without recalibration (20 subjects × 6 sessions, 64 channels). Rebuilt the organisers' scoring offline and evaluate on rotating cross-session folds that expose the drift a random split hides.

EXPERIENCE

SchoolBook.AI · Lead AI Architect

Jun 2025 – Jun 2026

- Built a **real-time voice AI tutor** that answers several students at once: raw PCM audio streamed over WebSockets into Gemini 2.5 Flash, **under 500 ms** from speech to action.
- Designed a fault-tolerant conversation state machine with voice-activity locks, so the tutor never answers into silence or talks over a student.
- Ran the audio services as containers handling many concurrent streams at **99.9% uptime**.
- Built a **multimodal RAG pipeline** (Gemini Pro and a vector database) that teaches from official NCERT textbooks, and **LangGraph agents** that plan lessons and use tools over several steps.

Qualcomm · Senior Lead Software Engineer

Apr 2023 – Dec 2024

- Cut GPU flicker-correction latency in the XR camera-rendering pipeline by **96% (35 ms → 1.3 ms)**, which made **90 Hz** refresh possible on premium Snapdragon chips.
- Designed new **HDR imaging flows** for premium-tier Snapdragon chips.

GreyscaleAI · Software Engineer (Computer Vision)

Feb 2021 – Sep 2022

- Architected the software that synchronises and fuses **X-ray and colour-camera** streams on a production inspection line for Nestlé, detecting foreign objects in real time.
- Built the real-time embedded UI (C++, Qt) that links the hardware sensors, the inference engine and the operators.
- Led a full codebase refactor that simplified the build for a team of 8 engineers.

- Career break** · Medical treatment and recovery Aug 2018 – Jan 2021
- Retrained from camera systems into deep learning during treatment: DeepLearning.AI's Deep Learning Specialization (2019) and TensorFlow Developer Certificate (2020).
- Nvidia** · System Software Engineer Jul 2017 – Jul 2018
- Fixed **lens-shading and Bayer-demosaicing** artifacts in the Tegra camera imaging stack used by computer-vision products.
- DRDO (Defence Research and Development Organisation)** · Scientist Aug 2013 – Jul 2015
- Built image enhancement and lens-shading (vignette) correction for **naval thermal imagers** with a **40 km** detection range, improving signal-to-noise on low-contrast thermal data.
 - Built a Raspberry Pi controller for the thermal imagers in-house, replacing an external purchase.

OTHER PROJECTS

Bella — a voice agent that takes part in live calls 2026
Python, faster-whisper, custom wake-word models, Gemini

- Joins a live Discord voice call, transcribes everyone continuously and answers aloud when called by name: **106 ms** to the AI's first word; local speech synthesis **27× faster** than real time.
- One speech recogniser per speaker (three people at once in 693 ms against 566 ms for one); two wake-word models ("Bella", "Stop") trained on about **10,000** recorded samples; the interruption check costs 0.08 ms per audio chunk.
- Solved what makes voice agents sound broken: hearing their own echo through other people's speakers, overlapping speakers, Discord's encrypted voice, and end-of-turn detection when 65% of call audio arrives as no packets. **373** automated tests.

SKILLS

AI and ML: PyTorch, TensorFlow, LLMs (Gemini, Claude, OpenAI), agents (LangGraph, LangChain), RAG, speech (Whisper), Google health models (MedGemma, HeAR, MedSigLIP)

Imaging and signals: Camera image pipelines (lens shading, demosaicing, HDR), 3D CT segmentation, remote photoplethysmography, EEG processing (braindecode, pyRiemann), spectral analysis, multi-sensor fusion

Engineering: Python, C++, FastAPI, Docker, Google Cloud Run, AWS, Linux, real-time audio streaming (WebSocket, PCM)

EDUCATION

IIT Delhi · M.Tech, Computer Technology 2015 – 2017

- Dissertation: thermal video stabilisation for night-vision systems, with DRDO.
- Project: EEG motor-imagery brain-computer interface for cursor control.

IIT Mandi · B.Tech, Electrical Engineering (founding batch) 2009 – 2013

- Founding member of the robotics section; built the institute's first robot. Dissertation: automobile collision-prevention system.

ACHIEVEMENTS AND COURSES

- GATE AIR 423 (99.9th percentile) · IIT-JEE top 1% · GRE 321/340 (Quant 170/170)
- Online: DeepLearning.AI Deep Learning Specialization (2019) and TensorFlow Developer Certificate (2020); Wharton Entrepreneurship Specialization (2021–22)
- Online: Johns Hopkins, Fundamental Neuroscience for Neuroimaging; Hebrew University, Synapses, Neurons and Brains; Northwestern, Modern Robotics